Cognitive Divergence and Convergence by Shared Alignment

Leibniz knew that each person has a unique individual perspective. In Gottfried Wilhelm Leibniz's major work, his "Monadology" from 1714, every monad acts as a unique, living view or perspective of the entire universe. Monads are unique and mirror the whole universe from their own point of view. Each monad contains an inner representation of the whole universe, a mirror of the cosmos. This is the unique individual perspective which is the reason for our subjective experience.

He also knew where this unique perspective came from. For Leibniz, the concept of an individual (or an individual substance which he discussed in his 1686 work "Discourse on Metaphysics") contains everything that ever happens to it. This is a metaphysical version of path-dependence where the state of an agent encodes its whole history. Since every person takes a slightly different path, every person has a unique individual perspective, and every person is unique. Leibniz' law, the identity of indiscernibles, states that says that two distinct things must differ in some property, and if two things share all their properties, they are the same thing.

Leibniz lived at a time where Germany was a big, fragmented patchwork of hundreds of independent states, including smaller kingdoms, duchies, principalities, and free cities. Each had its own dialect, currency and customs. Leibniz knew from experience that left to their own devices, people develop their own idiosyncratic ideas and unique subjective experiences. Their customs, habits, practices, languages and perspectives diverge.

But if people come together and share collective experience, they start to understand each other, and their perspectives converge. Emily Falk writes in her book "What we value" that neuroscientists have found out that "when two people's brains are in sync, they are more likely to understand each other" and "the brain patterns of close friends are remarkably similar to one another when making sense of the world". She writes "Importantly, when people watch and listen to the same media, spend time together, and communicate. their brains and bodies synchronize. By contrast, people who consume different media and have different life experiences diverge in how their brains respond to identical content".

For adaptive systems there is natural tension between

- convergence / coherence / organization
- divergence / decoherence / disorganization

Shared learning, common paths, and collective experiences lead to convergence of world views and mindsets. It pulls the agent's world views together, while individual learning, unshared paths, personal experiences lead to divergence and polarization of world views and mindsets, because they push the agent's world views apart.

Can we formalize this fundamental tension between convergence and divergence mathematically using path-dependent perspectives and probability theory?

Common sense says that the perspectives and subjective experiences of people must diverge, because they are adaptive agents that learn from experience through reinforcement. The average person meets roughly 50,000 - 80,000 people in a lifetime and knows words for about 60,000 objects. This is the size of a Chinese dictionary. The chance that two persons meet exactly the same set of persons in their life and valuate them in the same way is zero. The chance that two persons valuate the 60,000 objects we have in the world in the same way is also zero. The chance that two persons take the same path in their life is also zero, since two different persons cannot be at the same place at the same time, like two cars which cannot

be in the same parking place. A parking place can only be occupied once by a car. And since our perspectives and experiences are path-dependent and the paths differ, they must differ too.

The question is can we understand each other at all in principle or is it in principle not possible to understand perfectly what it is like to be someone else or what it is like to feel something. All signs point to the direction that a perfect understanding is not possible.

The FLP impossibility theorem in distributed computing proves that no deterministic consensus algorithm can guarantee both safety and liveness in a fully asynchronous distributed system if even a single process experiences a crash failure. We can try to formulate a similar impossibility theorem for the hard problem of consciousness. Let us name it the **LNC Impossibility Theorem**. It states that exact subjective equivalence between two conscious agents is impossible, and is named after Joseph Levine, Thomas Nagel, David Chalmers who first formulated the hard problem. It says in other words that exact subjective equivalence between two conscious agents is impossible and has probability zero, because each of them has a unique emotional valuation matrix which is the foundation of their subjective experience.

Why is subjective experience linked to emotions? Neuroscientist Antonio Damasio argues that consciousness began with feelings - the brain's translation of bodily states (like pain, hunger, or comfort) that tell an organism whether it is doing well or not. He has concluded that consciousness originates in interoception, the nervous system's internal monitoring of the body's physical state, rather than in abstract thought, language, or logic. If the connection between body and mind did not make us bad or good, it could not draw our attention to its presence of threats and opportunities (Damasio, 2026).

What do we mean by emotional valuation matrix? Emotional valuation matrix here means the system of value calculation in the ventromedial prefrontal cortex (vmPFC) that Emily Falk describes in her book "What we value" (Falk, 2025). She argues each person has a built-in value calculation system that converts many different considerations into a common, subjective sense of "how much this option is worth to me right now." Value in this context means subjective reward/value represented by the brain which reflects the assessment if it is good or bad for us. We can describe it as an emotional valuation matrix or graph (or as a matrix since every graph can be described by an adjacency matrix).

Because we are adaptive beings capable of learning, adjusting and changing this graph is path-dependent and depends on the history of a person. Subjective experience is based on the spectrum of this path-dependent graph. The more our paths diverge by living in different places and experiences different environments, the more these path-dependent graphs will differ. The more our paths converge, the more these path-dependent graphs will be similar too, if we learn the same lesson.

In real life the paths diverge from the first moment when we are born. As we grow older, we become both more uniquely subjective and more capable of transcending our subjective perspective (Glen's paradox of subjectivity mentioned in the first chapter). Just like two persons never have the same fingerprints, because their unique patterns are shaped by a combination of genetic instructions and random environmental factors, no two persons have the same "emotional valuation matrix". It is a similar combination of random fluctuations and growth which creates a unique non-ergodic path-dependent structure. As we say in probability theory the two structures are almost surely not isomorphic – which means the probability is 0, even if rare exceptions are mathematically possible.

The Shannon number is Claude Shannon's estimation of the complexity of chess: he estimated roughly 10^{120} possible games. A typical game has 40 moves. Claude assumed an average of about 10^3 possible move pairs (one White move and one Black move) per turn, and this gives over a typical game of about 40 moves: $(10^3)^{40} = 10^{120}$ possible games. Chess has 64 squares, 32 figures, and a few fixed rules. Imagine

how much larger the possible outcomes in real life are. Our adaptive brains reflect this complexity. Each of them is almost surely unique, since no two persons share the same path or experience the same world or learn the same things, and the brain contains ~ 10^{10} neurons which learn from experience. In addition, we have mechanisms such as preferential attachment that guarantee path dependence.

Why do we understand each other at all if we are so different? Because we have institutions that we are forced to visit that teach as the same languages and the same worldview. There are basically two types of forces that shape subjective experiences: random individual experiences naturally push people's worldviews apart over time so that they diverge, and shared collective experiences naturally pull people's worldviews together so that they converge.

If shared collective experiences outweigh the random influences, then subjective experience is converging and asymptotic approximation of subjective experience between two 1st-person views is possible. For pure subjective experience we get the **Asymptotic Qualia Theorem** (asymptotic approximation is possible)**.** It states that shared collective experiences naturally pull people's worldviews together. The more shared collective experiences they have, the more their world views will converge (if the coupling strength is stronger than the natural diffusion and divergence).

If random individual experiences are more frequent or stronger than shared collective experiences, then we can say that subjectivity is increasing and for pure subjective experience we get a **law of subjective divergence** (subjectivity is increasing), not unlike the 2nd law of thermodynamics which says entropy is increasing.

They can be considered as two corollaries A and B of the same law. We could name it the **fundamental theorem of worldview development** in complex adaptive systems. It says without shared forcing or collective experience by sharing the same path, random noise drives agents toward idiosyncratic isolation and increased subjectivity; with shared forcing and collective experience, active coupling pulls them towards shared worldviews. In a complex adaptive system of path-dependent agents, worldview divergence and alignment are two sides of the same coin.

Corollary A: random individual experiences naturally push people's worldviews apart over time. The less shared collective experiences they have, the more their world views will diverge. We can formulate it as a **law of subjective divergence**: it says the subjectivity increases, like the 2nd law of thermodynamics which says that entropy increases. Worldviews of uncoupled agents tend to drift apart.

Corollary B: shared collective experiences naturally pull people's worldviews together. The more shared collective experiences they have, the more their world views will converge (if the coupling strength is stronger than the natural diffusion and divergence). We can formulate this as a **shared alignment theorem**.

These theorems can be proved mathematically on different levels of abstraction using different theories. The proofs on the following pages are AI generated by Gemini and Claude. Since AI can be wrong and make mistakes, they must be taken with care. They serve as illustration that these theorems can be proved.

1. Setup and Assumptions

Let us say the worldview of agent *i* is represented as a conditional probability distribution $P(y \mid x; W_i)$ over a semantic space of about 60,000 dimensions, mapping environmental stimulus vectors x to subjective valuations y.

Agent i has parameters $W_i(t)$ in a d-dimensional parameter space, with predictive distribution $P(y \mid x; W_i)$. Fix a stimulus measure $p(x)$ (the distribution of situations the agents are compared on). Define the cognitive distance as the symmetrized Kullback-Leibler (KL) Divergence averaged over stimuli:

$$D(A||B) = E_{x\sim p}\left[KL\left(P_A(\cdot\,|x)\;||\;P_B(\cdot\,|x)\right) + KL\left(P_B(\cdot\,|x)\;||\;P_A(\cdot\,|x)\right)\right]$$

The measure p(x) must be specified explicitly; without it, an average over the entire stimulus space X is undefined.

Assumptions

These are modeling assumptions, not facts about minds. Everything below is a theorem about this model.

- **A1 (regularity):** $P(y \mid x; W)$ is smooth in W, all distributions share a common support, and the Fisher information matrix $F(W) = E_x E_y[\nabla_W \log P \cdot \nabla_W \log P^T]$ is finite.

- **A2 (identifiability):** Distinct parameters (modulo any discrete symmetry group, such as unit permutations) give distinct predictive distributions p-almost everywhere. We work in a gauge-fixed local chart where F is positive definite. This is what makes $W_A \neq W_B$ mean "worldviews differ".

- **A3 (local linear dynamics):** Both agents stay near the homeostatic attractor μ, so $F \approx F(\mu)$ is constant and the dynamics can be linearized around μ. Set $\mu = 0$ by translation.

- **A4 (nondegenerate noise):** Learning noise is additive Gaussian with $\sigma > 0$ in every direction, independent across agents. This is an idealization for agents exposed to the same stimulus stream: if their learning noise were positively correlated (correlation ρ), the noise on the difference would scale by $(1 - \rho)$ and the floor would shrink accordingly.

2. Lemma 1: Fisher-Metric Link Between Parameters and Distance

Let $\Delta = W_A - W_B$. Under A1, the standard expansion of KL divergence gives:

$$KL(P_A||P_B) = \frac{1}{2}\Delta^T F \Delta + O(||\Delta||^3)$$

and the same for the reverse direction. Summing the two:

$$D(A||B) = \Delta^T F \Delta + O(||\Delta||^3)$$

So cognitive distance is, to leading order, the squared Fisher-Rao length of the parameter difference. This links parameters to distance: any statement about $||\Delta||$ becomes a statement about D. Directions with large Fisher eigenvalue are "stiff" (small changes alter predictions a lot); sloppy directions barely matter.

3. Lemma 2: Dynamics of the Difference

We model each agent as an Ornstein-Uhlenbeck process with three ingredients: restoration toward the baseline state that the agents are pulled toward due to their shared biology (rate θ), optional pull toward a shared experience target or toward each other (rate k), and independent noise (σ).

- **Shared-experience model:**
  Both agents are pulled toward the same data-optimal parameters $W^*(t)$, because they see the same stimuli:

$$dW_i = \underbrace{-\theta W_i dt}_{\text{Restoring Force}} \underbrace{-k\big(W_i - W^*(t)\big)\,dt}_{\text{Shared Alignment Force}} + \underbrace{\sigma dB_i(t)}_{\text{Subjective Diffusion}}$$

  If we subtract the equation for two agents A and B

$$dW_A = -\theta W_A dt - k\big(W_A - W^*(t)\big)dt + \sigma dB_A(t)$$
$$dW_B = -\theta W_B dt - k\big(W_B - W^*(t)\big)dt + \sigma dB_B(t)$$

  The target W*(t) cancels exactly in the difference:

$$\begin{aligned} d(W_A - W_B) &= -\theta(W_A - W_B)dt - k[(W_A - W^*) - (W_B - W^*)]dt + \sigma(dB_A - dB_B) \\ &= -\theta\Delta\, dt - k\Delta\, dt + \sigma(dB_A - dB_B) \\ &= -(\theta + k)\Delta\, dt + \sigma(dB_A - dB_B) \end{aligned}$$

- **Mutual-coupling model (communication):**
  Agents are pulled toward each other by a mutual coupling or communication:

$$dW_A = \underbrace{-\theta W_A dt}_{\text{Restoring Force}} \underbrace{-k(W_A - W_B)dt}_{\text{Interaction Force}} + \underbrace{\sigma dB_A(t)}_{\text{Subjective Diffusion}}$$

  and symmetrically for B. Subtracting gives:

$$\begin{aligned} d(W_A - W_B) &= -\theta(W_A - W_B)dt - k[(W_A - W_B) - (W_B - W_A)]dt + \sigma(dB_A - dB_B) \\ &= -\theta\Delta\, dt - 2k\Delta\, dt + \sigma(dB_A - dB_B) \\ &= -(\theta + 2k)\Delta\, dt + \sigma(dB_A - dB_B) \end{aligned}$$

In the shared-experience model the common target cancels in the difference; in the mutual-coupling model the two interaction terms add rather than cancel (hence 2k). Distinct targets $W_A^* \neq W_B^*$ are treated in Law III. In both models, with independent noise, we get:

$$d\Delta = -\lambda\Delta dt + \sigma(\mathrm{dB_A} - \mathrm{dB_B}) = -\lambda\Delta dt + \sqrt{2}\sigma d\tilde{B}_\Delta$$

where $\tilde{B}_\Delta = (B_A - B_B)/\sqrt{2}$ is a standard Brownian motion with effective restoring rate $\lambda = \theta + k$ (shared experience) or $\lambda = \theta + 2k$ (mutual coupling). For the uncoupled case $k = 0$, $\lambda = \theta$.

This is a standard OU process, so with initial difference Δ_0:

- **Mean:** $E[\Delta(t)] = e^{-\lambda t}\Delta_0$
- **Covariance:** $Cov(\Delta(t)) = \frac{\sigma^2}{\lambda}(1 - e^{-2\lambda t}) \cdot I$

Combining with Lemma 1, the expected distance is (to leading order):

$$E[D(t)] = \epsilon + (a - \epsilon)e^{-2\lambda t}$$

where $a = \Delta_0^T F \Delta_0$ is the initial distance and:

$$\epsilon = \frac{\sigma^2 \mathrm{tr}(F)}{\lambda}$$

If the restoring rates differ by eigendirection (λ_k along the k-th Fisher eigenvector, with F and the restoring matrix simultaneously diagonalizable), the same computation gives the mode-wise version:

$$E[D(t)] = \sum_k f_k \left[\frac{\sigma_k^2}{\lambda_k}(1 - e^{-2\lambda_k t}) + \Delta_{0,k}^2 e^{-2\lambda_k t}\right]$$

where f_k are the Fisher eigenvalues.

Fluctuations

In the stationary regime, $D(t) = \Delta^T F \Delta$ is a weighted sum of squared Gaussians, with mean ϵ and variance $2\left(\frac{\sigma^2}{\lambda}\right)^2 \mathrm{tr}(F^2)$. With $d_{\text{eff}} = \frac{(\mathrm{tr}\ F)^2}{\mathrm{tr}(F^2)}$, the relative fluctuation is therefore:

$$\frac{\mathrm{std}(D)}{E[D]} = \sqrt{\frac{2}{d_{\text{eff}}}}$$

Here high dimension genuinely matters when many parameter directions contribute (large d_{eff}), D concentrates tightly around ϵ. This is a concentration-of-measure statement. Caveat: real Fisher spectra are often very sloppy, so d_eff can be much smaller than the nominal dimension, and the formula assumes a common σ and λ across directions.

4. Law I: LNC Impossibility Theorem - The Irreducible Cognitive Floor

Exact equivalence of two worldviews is almost surely not possible, because even when two agents share identical architecture, baseline attractor and experience stream, independent noise keeps their expected cognitive distance strictly above zero.

- **Statement:** Under A1–A4, for any $t > 0$, $P(D(A||B) > 0) = 1$.
- **Floor:** For any finite coupling ($\lambda = \theta + k$ or $\theta + 2k$), E[D(t)] ≥ ε(1 − e^(−2λt)) > 0 for every t > 0, and E[D(t)] → ε = σ² tr(F)/λ > 0 as t → ∞. If the agents start identical (a = 0), E[D(t)] rises from 0 toward ε; if a ≥ ε, then E[D(t)] ≥ ε for all t.

- **Proof:** By Lemma 2, for $t > 0$ the difference $\Delta(t)$ is Gaussian with mean $e^{-\lambda t}\Delta_0$ and covariance $\frac{\sigma^2}{\lambda}\left(1 - e^{-2\lambda t}\right)I$, which is full rank because $\sigma > 0$. A nondegenerate Gaussian has a density, so $P(\Delta(t) = 0) = 0$. By A2, $\Delta \neq 0$ implies the predictive distributions differ on a p-positive set, so $D > 0$.

- **Proof of the floor:** By Lemma 2, E[D(t)] = ε(1 − e^(−2λt)) + a e^(−2λt). The second term is non-negative, which gives the bound. The first term is strictly positive for t > 0 because σ > 0 and tr(F) > 0 (F is positive definite by A2).

What This Does and Doesn't Say

- The basic assumptions are noise (A4) and identifiability (A2), not spatial exclusion. Agents in disjoint bodies can receive identical stimulus streams; what keeps them apart in this model is independent noise.
- Real agents are finite-precision systems, so true coincidence has tiny but positive probability. The continuous model is an approximation, and the theorem should be read at that level. Note also that this is a measure-zero statement: it says nothing about how close two worldviews can come.
- **Expectation, not sample path:** The floor is a statement about E[D(t)]. A single D(t) fluctuates around ε (relative spread $\sqrt{2/d_{\text{eff}}}$) and can dip below it, but only rarely when d_{eff} is large.
- **Scope:** In the model the agents share architecture (F, θ), attractor μ and experience stream W*(t); only independent noise separates them. Path differences (different places, people and media) are not part of Law I; they enter through Law III, and the memory of older path differences through frozen modes or multiple basins (Section 5). If the two agents' noises are correlated (ρ < 1), the floor scales by (1 − ρ).

5. Law II: Convergence under shared experience and coupling

By sharing the same path and experience, the expected distance between worldviews relaxes to a positive floor (from above or below, depending on the starting distance).

- **Statement:** Under the shared-experience (or mutual-coupling) model with $\sigma > 0$, the expected distance converges exponentially to a strictly positive floor:

$$\lim_{t\to\infty} E\left[D(t)\right] = \epsilon = \frac{\sigma^2 \text{tr}(F)}{\lambda} > 0$$

  with $\lambda = \theta + k$ (shared experience) or $\theta + 2k$ (communication).

- **Sample Path:** The sample path $D(t)$ does not converge. It is a stationary process fluctuating around ϵ with relative spread $\sqrt{2/d_{\text{eff}}}$ (Section 3).
- **Proof:** Immediate from Lemma 2: $e^{-2\lambda t} \to 0$ for $\lambda > 0$.
- **Shared path:** Law II assumes the same target W*(t) for both agents, that is δ* = 0 (same places, media and people). Differences in paths are treated in Law III.

Consequences

- **The floor shrinks with coupling:** $\epsilon \propto \frac{1}{\theta+k}$.
  Stronger shared experience or communication lowers the floor, but it reaches zero only when noise does ($\sigma \to 0$) or coupling is infinite.
- **Convergence time:**
  The approach to the floor has time scale $\frac{1}{2\lambda}$. Tighter coupling both lowers the floor and gets there faster.

Priors and Permanent Offsets

Priors are forgotten in this model. The initial offset enters only through $(a-\epsilon)e^{-2\lambda t}$. A permanent offset from prior histories is therefore not a consequence of OU dynamics. It needs extra structure, any of which is a legitimate modeling choice:

1. **Frozen or very slow modes:** If some eigendirections have $\lambda_k = 0$ and $\sigma_k = 0$ (developmental parameters that stop changing), the offset persists exactly: those modes contribute $f_k \Delta_{0,k}^2$ to ϵ. If λ_k is merely small, the offset persists on time scales shorter than $\frac{1}{2\lambda_k}$ which for a human lifetime may be effectively permanent. Note that this relaxes A4 for those directions ($\sigma_k = 0$). Law I still holds, since $\Delta_k = \Delta_{\{0,k\}} \neq 0$ deterministically.

2. **Multiple basins:** A non-convex landscape with several attractors means agents that start in different basins do not merge, however much experience they share.

3. **Pure random walk:** A mode with $\lambda_k = 0$ and $\sigma_k > 0$ is a pure random walk and diverges (Section 6).

With frozen modes F_{frozen}, the general floor is:

$$\epsilon = \sum_{k:\lambda_k>0} f_k \frac{\sigma_k^2}{\lambda_k} + \sum_{k:\lambda_k=0,\sigma_k=0} f_k \, \Delta_{0,k}^2$$

The two terms correspond to two mechanisms: stochastic noise and prior histories.

6. Law III: Divergence through path difference

Agents that live in different places and follow different paths are pulled toward different targets. The expected distance then contains a systematic path term on top of the noise floor. The path term grows with the path difference and, if the paths keep drifting apart, without bound, even though the restoring force is present.

- **Statement:**
  Let each agent be pulled toward its own stream target, $W^*_{A(t)}$ and $W^*_{B(t)}$), at rate k > 0, and let $\delta^*(t) = W^*_{A(t)} - W^*_{B(t)}$ be the path difference (different places, people, media).

  Then $d\Delta = -\lambda\Delta\, dt + k\,\delta^*(t)\, dt + \sigma(dB_A - dB_B)$, with $\lambda = \theta + k$, and the expected distance is $E[D(t)] = \varepsilon\left(1 - e^{-2\lambda t}\right) + m(t)^T F\, m(t)$, where $\varepsilon = \sigma^2\, tr(F)/\lambda$ and $m(t) = e^{-\lambda t}\,\Delta_0 + k \int_0^t e^{-\lambda(t-s)}\, \delta*(s)\, ds$. The first term is the noise floor of Law I; the second is the path term.

- **Proof:**
  Subtract the two OU equations as in Lemma 2 but keep the targets: $-k(W_A - W_A^*) + k(W_B - W_B^*) = -k(\Delta - \delta*)$, so the drift of Δ is $-(\theta + k)\Delta + k\delta^*$.
  The noise is unchanged, so $\mathrm{Cov}\,\Delta(t) = \left(\frac{\sigma^2}{\lambda}\right)\left(1 - e^{-2\lambda t}\right) I$ and $E[\Delta(t)] = m(t)$.
  Then $E[D] = \mathrm{tr}(F\,\mathrm{Cov}\,\Delta) + m(t)^T F\, m(t)$.

- **Shared path ($\delta^* = 0$):** The path term decays to zero and Law II is recovered: only the noise floor remains.

- **Constant path difference:** $m(t) \to (k/\lambda)\,\delta^*$, so $E[D(t)] \to \varepsilon + (k/\lambda)^2\,\delta^{*T} F\,\delta^*$. Different paths raise the plateau by a systematic term.

- **Drifting paths:** If $\delta^*(t)$ wanders (for example a random walk with covariance $\rho^2 t I$), then for $t \gg 1/\lambda$, $m(t) \approx (k/\lambda)\,\delta^*(t)$ and the path term grows approximately like $(k/\lambda)^2 \rho^2\,\mathrm{tr}(F)\,t$, that is linearly, even though $\theta > 0$. If the paths stay confined, the path term is bounded: by Jensen's inequality, for $t \gg 1/\lambda$, $m(t)^T F\, m(t) \le (k/\lambda)^2$ max over $s \le t$ of $\delta^*(s)^T F\,\delta^*(s)$.

- **Corollary (non-coincidence without noise):** Set $\sigma = 0$ and $\Delta_0 = 0$ (identical initial worldviews). If δ^* is constant and nonzero and $k > 0$, then $D(t) = (k/\lambda)^2 (1 - e^{\wedge}(-\lambda t))^2\,\delta^{*T} F\,\delta^* > 0$ for every $t > 0$, because F is positive definite (A2). For a time-varying $\delta^*(t)$, $D(t) = 0$ requires $m(t) = 0$, that is, a weighted average of the past path difference must vanish exactly, which is non-generic. So coupled agents on different paths have different worldviews even without noise. Since no two embodied agents occupy the same place at the same time, their paths always differ; this is the sense in which equivalence would require identical paths.

- **Relation to Leibniz:** Leibniz's identity of indiscernibles says that two distinct things cannot share all their properties. For worldviews, the corollary is a dynamical version of this principle: in the noise-free model, coinciding worldviews require the weighted path difference m(t) to vanish, and for two embodied agents it does not. The premise doing the work is that no two embodied agents share a complete path. A world with two perfectly symmetric agents would violate that premise, and the result would no longer follow. Unlike the metaphysical principle, the result is graded: it gives a lower bound on the distance, not only a yes/no verdict.

- **Role of the coupling:** Coupling to one's own stream lowers the noise floor $\varepsilon = \sigma^2\,\mathrm{tr}(F)/(\theta + k)$ but raises the sensitivity to the path difference, $(k/(\theta + k))^2$. For large k, $E[D] \to \delta^{*T} F\,\delta^*$: each agent tracks its own stream, and the distance between the minds equals the distance between their data-optimal worldviews. If the agents are also pulled toward each other at rate k_c, then $\lambda = \theta + k + 2k_c$, the forcing is still $k\delta*$, and the plateau becomes $\varepsilon + (k/\lambda)^2\,\delta*^T F\,\delta*$; mutual coupling lowers both terms.

- **Limits:** With $\lambda > 0$ the model forgets path differences older than about $1/\lambda$. Findings that people with different life experiences respond differently to identical content suggest slow or frozen modes (Section 5), which would keep older path differences in the distance.

Special case: no coupling ($k = 0$). With $k = 0$ the stimulus stream does not enter the dynamics at all, so the plateau comes from noise alone: "unshared paths" matter only through $k > 0$ as above.

- **Statement:** For uncoupled agents ($k = 0$) with independent noise and restoring rate $\theta > 0$:

$$\frac{d}{dt}E[D(t)] = 2e^{-2\theta t}(\sigma^2 \mathrm{tr}(F) - \theta a)$$

where $a = \Delta_0^T F \Delta_0$. Hence:

- $\frac{d}{dt}E[D] > 0$ if and only if the initial distance a is below the plateau $\epsilon_0 = \frac{\sigma^2 \mathrm{tr}(F)}{\theta}$, which includes the common case where the agents start close ($a < \epsilon_0$).
- Starting above the plateau, $E[D]$ decreases toward ϵ_0.
- In all cases $E[D(t)] \to \epsilon_0 = \frac{\sigma^2 \mathrm{tr}(F)}{\theta}$, a finite value

- **Proof:** Differentiate $E[D] = \epsilon_0 + (a - \epsilon_0)e^{-2\theta t}$ from Lemma 2: the derivative is $-2\theta(a - \epsilon_0)e^{-2\theta t}$ which equals the expression above.

Where Unbounded Divergence Does Occur

If the restoring force is absent ($\theta = 0$ and $k = 0$), Δ is a pure random walk with $Cov(\Delta) = 2\sigma^2 t \cdot I$, and:

$$E[D(t)] \approx a + 2\sigma^2 \mathrm{tr}(F)t$$

grows linearly, if the quadratic approximation (Lemma 1) holds. With independent noise alone, unbounded divergence therefore requires that "shared human hardware" exerts no restoring force. The OU model above says the opposite, since it includes $-\theta(W - \mu)$.

Drifting paths (see above) give a second route to unbounded growth even when θ > 0.

7. Summary of the Laws

The laws describe formally that noise and random encounters keep people from being identical, shared experience pulls them together, and different paths pull them apart.

Law	Meaning	Exact Claim	Needs
I	Distance $D > 0$ for two agents	$D > 0$ almost surely for $t > 0$; E[D(t)] ≥ ε(1 − e^(−2λt)) > 0	$\sigma > 0$, identifiability
IIa	Expected distance relaxes under shared experience to $\lim D = \epsilon > 0$	$\lim E[D] = \frac{\sigma^2 \mathrm{tr}\, F}{\theta + k} > 0$; $D(t)$ fluctuates around it	$\sigma > 0$ for a positive floor; permanent prior offsets additionally need frozen modes or multiple basins
IIb	Expected distance relaxes under mutual coupling to $\lim D = \epsilon > 0$	$\lim E[D] = \frac{\sigma^2 \mathrm{tr}\, F}{\theta + 2k} > 0$; $D(t)$ fluctuates around it	$\sigma > 0$ for a positive floor; permanent prior offsets additionally need frozen modes or multiple basins
III	Path difference. Agents on different paths ($\delta^* \neq 0$) are pulled apart: distance has a path term on top of the noise floor.	E[D(t)] = ε(1 − e^(−2λt)) + m(t)^T F m(t); plateau ε + (k/λ)² δ*^{T} F δ* for constant δ*; linear growth for drifting paths; D > 0 even without noise if δ* ≠ 0	k > 0 (pull toward own stream); δ* ≠ 0; drifting δ*(t) for unbounded growth
III (k = 0)	No coupling. Without coupling, expected distance	$E[D]$ relaxes monotonically to	$\theta > 0$ for plateau; $\theta = 0$ for unbounded drift

	relaxes to a plateau, rising only when a < ε₀: $\frac{d}{dt}E[D] > 0$	$\frac{\sigma^2 \text{tr}\, F}{\theta}$; grows linearly only if $\theta = 0$	

Laws IIa, IIb and III are special cases of one formula:

$$E[D(t)] = \epsilon + (a - \epsilon)e^{-2\lambda t}$$

with $\epsilon = \frac{\sigma^2 \text{tr}(F)}{\lambda}$.

- Law I says $\epsilon > 0$ and $D > 0$ almost surely.
- Law II says coupling raises λ.
- For k = 0, Law III says that with θ small, ϵ is large and the approach to it is slow. Law III (paths) adds that different paths contribute a systematic term $(k/\lambda)^2\, \delta^{*T} F\, \delta^*$, which grows without bound if the paths keep drifting apart.

8. What Remains an Assumption, and What Would Test It

The mathematics above is internally consistent. Whether it applies to minds depends on modeling choices that the framework does not justify on its own:

- That a worldview is well modeled by a smooth conditional distribution $P(y \mid x; W)$ with a finite-dimensional W, and that "subjective valuation" is the output y.
- That learning noise is additive, Gaussian, and independent across agents, and that it is not state-dependent.
- That the dynamics are linearizable around a single biological attractor μ (A3).
- That the relevant stimulus measure p is well defined.
- That an agent's experience enters as a target W*_i(t) with a linear pull of rate k, and that the path difference δ*(t) can be measured (places, people, media).

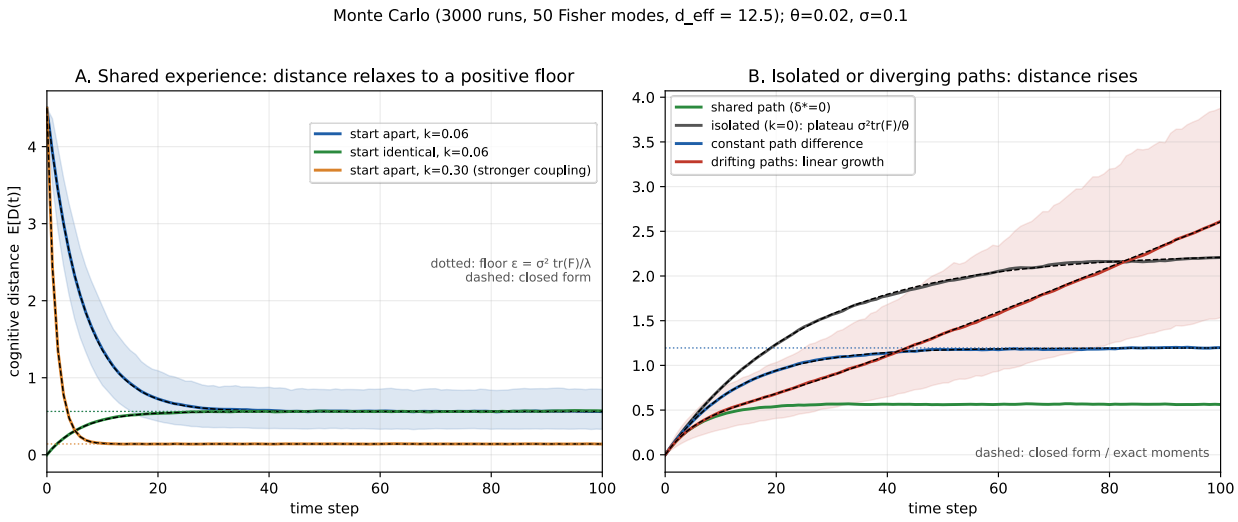
The revised model makes quantitative, falsifiable predictions, if D can be operationalized (for example, as divergence between two people's rating distributions on the same stimulus set):

1. **Plateau, not growth:** Pairs separated for longer should show distance that levels off rather than increasing without bound.
   This holds only if the paths do not keep drifting apart (see Law III).
2. **Floor scaling:** The floor should scale inversely with the strength of shared experience or communication, $\epsilon \propto \frac{1}{\theta + k}$.
   For mutual coupling the denominator is θ + 2k.
3. **Relaxation time:** Convergence between newly coupled pairs should be exponential, with time scale $\frac{1}{2\lambda}$.
   This is the time scale of E[D]; Δ itself relaxes at rate λ.
4. **Concentration:** When many independent dimensions contribute, pair-to-pair variability of D around its mean should shrink like $\frac{1}{\sqrt{d_{\text{eff}}}}$.

5. **Persistent offsets:** Residual distance at long times beyond the noise floor would indicate frozen modes or multiple basins, and its size measures them.

6. **Path dependence:** Beyond the noise floor, distance should increase with the difference in two people's paths (places, people, media) through the term $(k/\lambda)^2\ \delta^{*T} F\ \delta^*$, and fall as their paths overlap more. Findings reported in the neuroscience of inter-subject synchrony (for example as summarized in Falk, What We Value) point in this direction, but they should be tested against this model quantitatively.

If data show unbounded growth, the restoring term θ should be dropped or the dynamics made non-stationary. If data show convergence to zero between tightly coupled pairs, ϵ must be driven to zero, which means noise must be negligible. Either outcome is informative. To keep the model refutable, θ, k and σ should be estimated independently (for example from within-person test–retest drift and from pairs with known exposure), so the plateau and floor are predictions rather than fits.



Left: disagreement converges by shared experience to a positive floor
Right: disagreement of isolated agents grows linearly; with coupling it stays at the floor.

The minimal distance (ϵ)

The floor discussed above is only for comparing the same species. World views based on emotional valuation matrices can only be recreated for sensory organs which uses similar receptors that return similar results. If the sensory organs or receptors are completely different, it does not make sense to compare valuations based on it.

For example, humans have 180-degree vision and three color receptors, while horses and sheep have 350-degree vision and only two color receptors, and whales and dolphins have monochromatic vision because they only have one color receptor. For bats and whales specialized hearing replaces vision in dark environments, for fish feeling by the lateral line system replaces sound, and for birds and insects perceiving the magnetic fields helps in navigating complex environments.

Here is a table comparing the sensory organs and receptor similarities between humans and various species. The similarity index ranges from 0 (completely absent or fundamentally different evolutionary origin) to 1 (virtually identical structure and function), with higher values reflecting shared vertebrate or mammalian traits.

Sensory Receptors & Organs Comparison Table

Sensory Organ / Receptor Type	Mammals (Humans, Apes)	Mammals (Horses, Sheep)	Mammals (Whales, Bats)	Reptiles & Birds	Fish	Insects
Pain Receptors (Nociceptors)	1.0	1.0	1.0	1.0	0.9	0.8
Taste Receptors (Gustatory)	1.0	1.0	1.0	1.0	0.7	0.4
Binocular & Color Vision	1.0	0.8	0.8	0.8	0.8	0.3
Hearing (Auditory / Mechanoreceptors)	1.0	1.0	0.8	0.8	0.5	0.4
Olfactory Receptors (Smell)	1.0	1.3	0.6	0.9	0.8	0.9
Lateral Line System	0.0	0.0	0.0	0.0	1.0	0.0
Magnetoreception	0.0	0.0	0.2	0.8	0.5	0.7

We can use this table to formulate a distance metric ϵ, where we use each species' sensory profile as a vector in a 7-dimensional space representing the 7 sensory receptor/organ categories and calculate a **Normalized Euclidean Distance** or **Dissimilarity Index** relative to the human baseline.

Dissimilarity Values and Calculated minimal distance (ϵ)

Let vector H represent the Human sensory profile, and vector S represent the target species profile. We define the dissimilarity metric $\epsilon(H, S)$ as:

$$\epsilon(H,S) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(H_i - S_i)^2}$$

Where $n = 7$ (the number of sensory systems evaluated). This computes the root-mean-square deviation between the human baseline and each species' scores, naturally bounding and scaling the results to match your intuitive progression. Applying this formula to our comparison table yields the following progressive dissimilarity gradient from humans:

Species Comparison	Target Group	Calculated ϵ Score	Scaled (*2)
Human vs. Human	*Self-comparison*	0.00 +/- 0.02	4% ~ 0.04
Human vs. Mammal (Horse/Cow)	*Close Terrestrial Mammals*	0.14	28% ~ 0.28
Human vs. Mammal (Whale/Bat)	*Specialized Mammals*	0.20	40% ~ 0.40
Human vs. Reptile / Bird	*Distant Vertebrates*	0.32	64% ~ 0.64
Human vs. Fish	*Aquatic Vertebrates*	0.49	98% ~ 0.98
Human vs. Insect	*Invertebrates*	0.50	100% ~ 1.00

The Evolutionary Core of Qualia

What can we reply to sceptics who still argue that all this (even the most complicated theorems and sophisticated calculations) do not solve the hard problem and that's not the hard problem they were asking about?

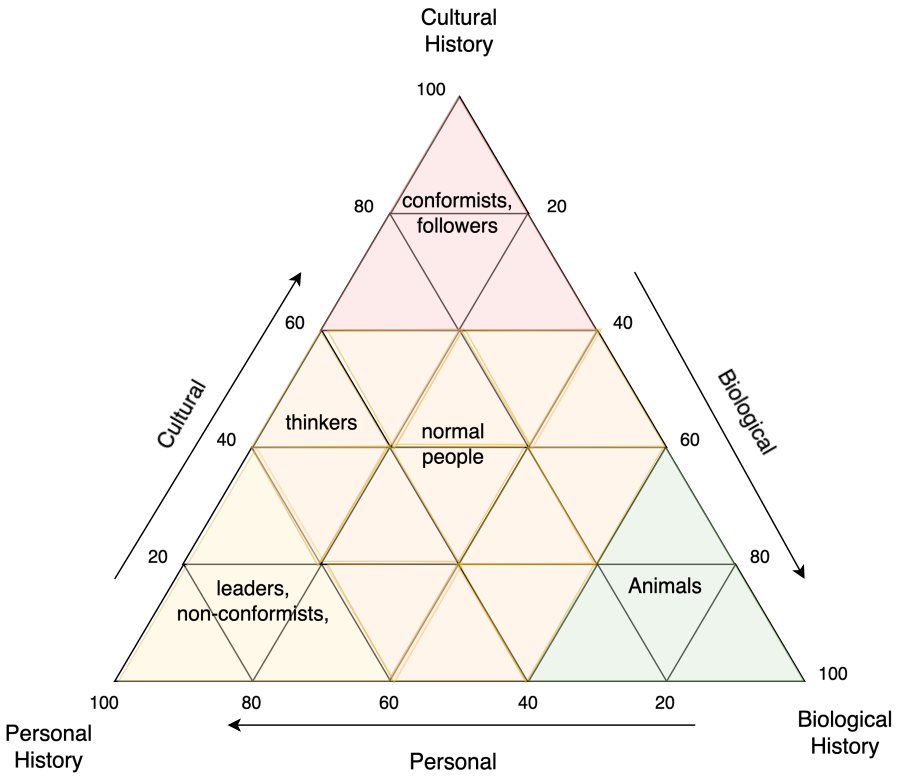
We can divide subjective experience into different layers which depend on biological, cultural and personal path-dependence:

Layer	Depends on	Solvable/Reducible	Notes
Physical / Biological	Molecular biology and genes: receptors for pain and pleasure, for red, green and blue colors, for sweet, salty, sour and bitter taste. Species-specific genes. Biological path-dependence.	For different species: no For same species: understandable in principle	Understandable in principle by another person of the same species (because the receptors are species-specific), but not in detail
Cultural	The path-dependent history of a culture. Cultural path-dependence.	Yes, approximately by following and understanding the path and the history of the culture if it is known	The Sapir-Whorf hypothesis says that the structure of a language shapes and influences its speakers' worldview, cognition, and perception of reality. In China for example the colors white and red have different meaning than in Europe.
Personal	The path-dependent history of a person. Personal path-dependence	Yes, approximately by another person of the same species following the same path by transforming different 1st-person experiences	Absolute equivalence and understanding of subjective experience is not possible

To understand the biological path-dependence we need to look back far into history and examine how species have developed and why animals are built the way they are. This gives us an understanding why emotional valuations or emotions in general exist and what their purpose is.

Cultures and nations are much younger than biological species – typically only a few hundred years. To understand cultures, we need to look at their history too. To understand individual people, we only need to know their brief lifetime which is less than a hundred years.

For animals the cultural influence is minimal, while for leaders and non-conformists the cultural influence is minimal. As a ternary chart it looks like this:



There are several aspects and dimensions of the explanatory gap that Joseph Levine has observed: biological vs cultural vs personal, 1st vs 3rd person view, knowing vs imagining, and what vs why.

The first dimension is the point of view: can we explain the difference between 1st and 3rd person views (why does the 1st person view feeling like something at all, while the 3rd person view is not connected to personal feelings), and can we find an translation between two 1st-person perspectives.

The second dimension is difference between knowledge and imagination:

1. knowing what it is like to be someone and to feel something
2. imagining what it is like to be someone and to feel something

The third dimension is to distinguish between the “what it is like” question (what it is like to be someone or what it is like to feel something) and the “why is it like” question (why do we feel something at all or why does it feel the way it does if we experience something).

The ultimate answer to the “why” question can be answered by evolutionary psychology, biology and ethology. The 1st person view is connected to personal feelings and the 3rd person view is not because subjective experience is always linked to a subject, and a subject is always linked to feelings and emotions which tell it what to do - unless it is a robot or a large language model.

The “what it is like” question is the one that matters pragmatically in our lives. The “knowing what it is like” cannot be solved perfectly. If we cannot understand subjective 1st-person experience in an objective way by measuring it, can we at least try to understand it by transforming a 1-person experience from one person to another? The answer to this is - as the theorems try to show - that exact equivalence of subjective experience is unattainable, and shared experience gets you only to a floor that is almost surely

above a zero distance. But even this minimal understanding gives us access to the larger question. The “imagining what it is like” question can be solved approximately.

As members of the same biological species, we know what it is like to be a human being that experiences pleasure and plain and all the other basic emotions. Each of us knows what it is like to feel good or bad, happy or sad. The longer we see what the other person went through, the more we understand what it must be like to be that person.

Explaining qualia is like the conquering an ancient, impregnable castle which is protected by many walls and layers. The walls have no windows, and we cannot see what is going on inside. The central keep of the castle is unreachable. But we can conquer the outer walls. If we try long enough, we get a glimpse what is going on inside and why the walls are built the way they are. The outer walls depend on the history of the castle: the inner buildings were built first, and the outer layers evolved later.

The outer walls for the hard problem depend on the history of a person. If we mean by the 'Hard Problem' understanding of what it is like to be someone or what it is like to feel something, then this can be solved approximately for two persons. The more the two share the same path and learn the history of the other person, the better they understand each other. If they have the same physiology, i.e. the same genetic setup and the same receptors they have a chance to understand each other. This is the hard problem that matters, as Daniel Dennett would say. It can be solved approximately by asymptotic approximation.

The inner walls and the central keep are harder to reach. The central keep of a castle is built on the bedrock below. If we mean by the 'Hard Problem' explaining why physical processes are accompanied by a certain kind of experience at all, then this bedrock question can be reduced and boiled down to the question why a biological body built for survival experiences valence (pleasure/pain) as a directional force. The answer is embodied homeostatic and ethological necessity — the irreducible dimensions of qualia reflect the orientation vectors of physical self-preservation.

Pain feels bad because it is signal of a survival alarm system designed to protect your body from damage. It indicates events or forces that are harmful for the body and damage its physical integrity. We can say pain is not only a message of a survival alarm system, it is the alarm system. *Pain feels like a message from an alarm system because it is an alarm system built to protect the body*. It is the alarm system in action that regulates our behavior where the genes criticize our behavior and warn about damages. As Nicholas Thompson said emotions can be seen as regulations of behavior (see the first chapter). He argued “pain is an emergency organization of my behavior in which I deploy physical and social defenses of various sorts”.

Pleasure is the opposite of pain. It feels good not only because it is signal of our control system designed to reward and encourage further action, it is the reward system in action. *Pleasure feels like a message from a reward system because it is a reward system built to reward the actions that benefit the body*. It is the reward system in action that regulates our behavior where the genes praise our behavior. We can view pleasure is a organization of behavior where we are told “yes, you are on the right track. go on. do more of this. it is going well. continue, engage, consolidate this. do this. you are in alignment with your primary directive”. The is the message of pleasure which our genes whisper if we are on the right track.

The answer to the first question (what it is like to feel something at all and why structure is accompanied by experience in the first place) can therefore be answered by everybody personally. This answer is in my opinion ultimately anchored in our biological nature. Our minds are inextricably linked to a single, unique, biological body. Our genes have created biomolecular control mechanisms - emotions - to control us. They tell us in every moment if something is good or bad, if something should be approached or not, if something should be consumed or not. These are the irreducible dimensions of subjective experience.

They are linked to the needs of our biological body - mainly survival and reproduction. Whenever something is happening in nature in the biological world it is either supper or pairing time. The most basic emotions hunger and disgust (consume it-expel it), pleasure and pain (do it-stop it), love and fear (approach it-get away from it) are direct diametrical opposites of primary subjective dimensions.

Irreducible Core Dimension	**Biological / Physical Meaning**	**Operational Meaning / Orientation Vectors**
Pleasure vs. Pain	Tissue Maintenance vs. Tissue Damage	"Engage with this / Yes, do this" vs. "Protect yourself / No, stop this"
Hunger vs. Disgust	Energy Intake vs. Toxicity Avoidance	"Incorporate into Self" vs. "Expel from Self"
Love/Attraction vs. Fear	Reproduction vs. Threat Evasion	"Approach / Merge" vs. "Flee / Run away"

Thus, if we try to find out what it is like to feel something we eventually will hear behind the whole bio-molecular machinery of our emotions the whisper of our primary biological directive that is encoded in our genes. They are the ultimate judges if something should feel good or bad.

Pleasure and pain are the most fundamental irreducible dimensions of subjective experience. They feel like they do because we have the bodies we have, which are built for survival and experience emotional valuations to guide our behavior. If we mean by the 'Hard Problem' explaining why a biological body built for survival experiences valence (pleasure/pain) as a directional force, then the answer is embodied homeostatic necessity — the irreducible dimensions of qualia are the physical orientation vectors of self-preservation.

Summarized in one sentence we can say in the tradition of Antonio Damasio, Randolph Nesse, and Nicholas Thompson: *The irreducible core dimensions of qualia (what things feel like) reflect the orientation vectors of physical self-preservation. Forged by evolution, these valenced states instantly steer behavior toward survival.*

If we want to understand the deepest inner mystery of subjective experience, and the inner keep of the castle, then I am convinced this is the answer we will find. Pleasure is pleasing for me for the same reason why pleasure is painful for you: it is the reward system in action which was built to reward our bodies. People who have the same physiology have the same reward system: the same dopamine molecules, the same neurotransmitters and the same basic pathways and circuits. Therefore, the experience of this irreducible core dimension must be similar among all beings that have the same physiology.

The same argument can be made for pain. Pain is painful for me for the same reason why pain is painful for you: it is the alarm system in action which was built to protect our bodies. People who have the same physiology have the same alarm system: the same A-delta and C-fibers, the same pathways. Again we can argue that the experience of this irreducible core dimension must be similar among all beings that have the same physiology.

The time when these systems become active exactly - or how strongly they do it - varies from person to person, but we basically all have the same system. If we all have the same alarm system at home, we will experience a similar sound if the alarm goes on. It might be configured a little bit different. As Nicholas Thompson wrote "I tend to think of emotions as regulations. What makes them vary among people is that people live in different worlds (well, different "slices" of the same world) and have different regulator set points, determined both by their experience and innate physiology" (see first chapter).